

Motivation and Question

- AI labs measure econ. attributes of AI usage data w/ privacy-preserving **classifiers**.
 1. Coding task “complexity” (Johnston et al., 2026): gpt-5-mini given Codex sessions w/ prompt to estimate “active work time an experienced human would need” to complete task.
 2. Task value (Hitzig et al., 2026): format session’s core task as freelance job posting, use Claude Opus 4.7 to estimate its price.
- **Measurement error?** $\implies$ researchers check this on **benchmark** datasets.
 1. Johnston et al. (2026): 1000 Codeforces problems $\implies \text{cor} = 0.65$ between gpt-5-mini and actual competitive coding times.
 2. Hitzig et al. (2026): 999 freelance job postings selected w/ stratified sampling on price tiers $\implies R^2 = 0.38$ between log actual and log Claude predicted price.
- What is the right metric? How wrong is estimate of avg. task value θ ? Can we correct it?

Answers from Econometric Theory

- **Missing data** problem: privacy $\implies$ never see truth in usage... but see it in benchmark!
- Idea: compute **classifier error in benchmark** and use it to adjust usage estimate, accounting for **distribution shift** (benchmark not perfectly representative of usage).
- Econometric theory (Chen et al., 2008; Carlson and Dell, 2025) $\implies$ measurement error corrected estimator should be: Y =value in benchmark, X =session task info, $\text{GPT}(\cdot)$ =classifier,

$$\hat{\theta}^{\text{old}} = \underbrace{\frac{1}{|\text{usage}|} \sum_{i \in \text{usage}} \text{GPT}(X_i)}_{\text{simple avg. of predictions}}, \quad \hat{\theta}^{\text{new}} = \hat{\theta}^{\text{old}} + \underbrace{\frac{1}{|\text{bench}|} \sum_{i \in \text{bench}} \underbrace{\hat{w}(X_i)}_{\text{re-weight due to shift}} \underbrace{\{Y_i - \text{GPT}(X_i)\}}_{\text{error}}}_{\text{bias correction term}}.$$

- $\hat{w}(\cdot)$ given by theory; requires estimating prob. task X was in usage (v. benchmark) data.
- $\hat{\theta}^{\text{new}}$ unbiased for **true θ** we care about (under sufficient conditions, in large samples).

Takeaways for Econ. Teams at AI Labs

1. Theory very **general** $\implies$ debias estimate of any θ (e.g., linear reg. coefficient.).
2. Choose benchmark dataset w/ tasks as **similar** as can be to those in usage population.
 $\rightarrow$ Johnston et al. (2026): **curate** dataset of usage-like coding tasks done by non-comp. coders.
3. **Only one more** thing to engineer: privacy-pres. **classifier** of usage v. benchmark data.
4. Core distribution shift assumption violated when benchmark **selected on outcome**.
 $\rightarrow$ Hitzig et al. (2026): should stratify on price *predicted* from task X instead of actual price Y .
5. **“Strong”/expensive** model can also be used as a ground truth signal (w/ caution!).

Econometric theory has a lot to say about AI evaluation!

References I

Carlson, Jacob and Melissa Dell (2025). *A Unifying Framework for Robust and Efficient Inference with Unstructured Data*. Version Number: 2. DOI: 10.48550/ARXIV.2505.00282. URL: <https://arxiv.org/abs/2505.00282> (visited on 09/03/2025).

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Hitzig, Zoe et al. (June 16, 2026). *Agentic coding and persistent returns to expertise*. URL: <https://www.anthropic.com/research/research/claude-codeexpertise>.

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